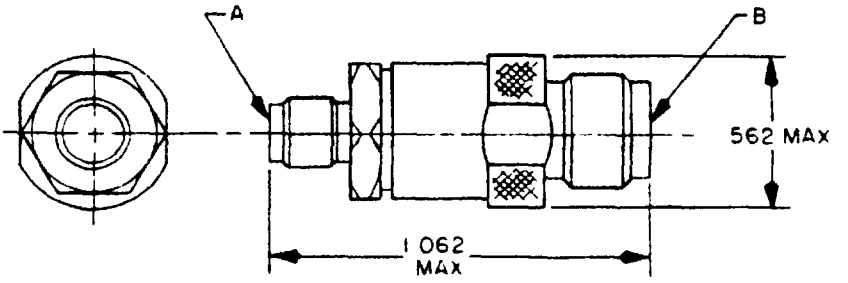
MILITARY SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA (FEMALE) TO SERIES TNC (FEMALE)), CLASS 2, STRAIGHT PLUG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapter described herein shall consist of this document and the latest issue of Specification MIL-A-55339



Reference	Series	Contact	Figure
A	SMA	Socket	2
B	TNC	Socket	3

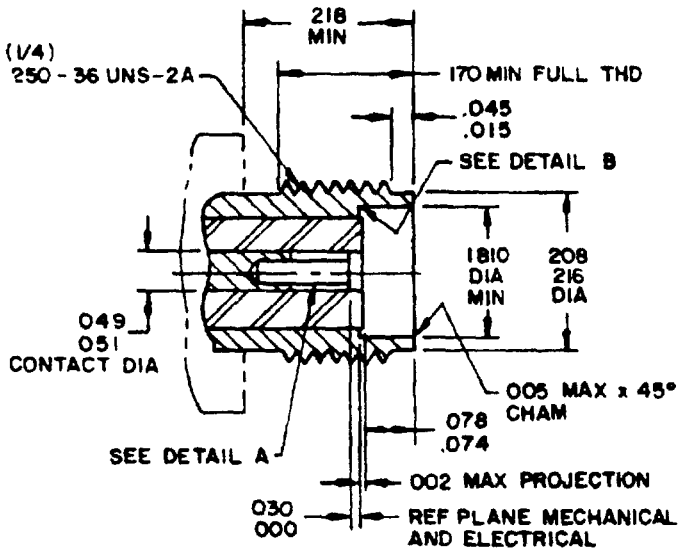
INCHES	MM
562	14 27
1 062	26 97

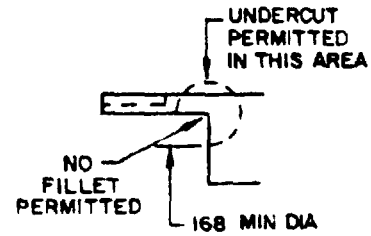
NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3 All undimensioned pictorial representations are for reference purposes only
4 Wrench flat to accommodate standard wrench per H-28, appendix 10

FIGURE 1 General configuration

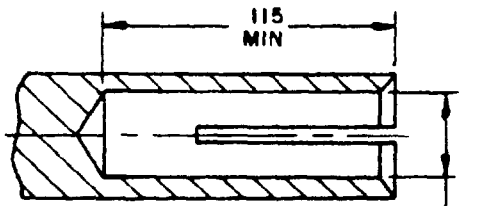
FSC 5935

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DETAIL B



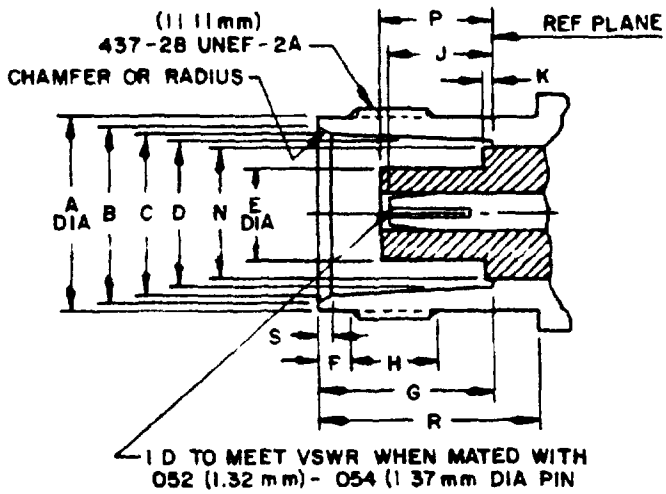
DETAIL A

INCHES	MM
002	05
003	08
.005	13
015	.38
.030	76
.0355	902
0370	940
045	1 14
.049	1 24
051	1 30
.074	1.88
078	1 98
115	2 92
170	4 32
1810	4 597
208	5 28
216	5 49
.218	5 54
250	6 35

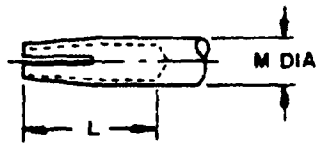
NOTES
1. Dimensions are in inches
2. Slitting of inner contact optional
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm

FIGURE 2 Mating dimensions for socket termination

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)			
	Minimum		Maximum	
A	.378	(9.60)	.381 (9.68)	
B	.345	(8.76)	.356 (9.04)	
C	.327	(8.31)	.333 (8 46)	
D	.319	(8.10)	.321 (8.15)	
E			.186 (4.72)	
F	.068	(1.73)	.088 (2.24)	
G	.329	(8.36)	.333 (8.46)	
H	.187	(4.75)		
J	186	(4 72)	.206 (5.23)	
K			.006 (.15)	
L	.195	(4.95)		
M	.081	(2.06)	.087 (2.21)	
N			.256 (6.50)	
P	188	(4 78)	.208 (5.28)	
R	.415	(10.54)		
S	.015	(.38)	.030 (.76)	



*N dimension applies to that portion (if applicable) of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension K

NOTES
1 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm.
2 All undimensioned pictorial configurations are for reference purposes only

FIGURE 3 Mating dimensions for socket terminations.

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ENGINEERING DATA

Nominal impedance 50 ohms.
Frequency range 0 to 11 GHz.
Voltage rating 335 volts rms maximum working voltage at sea level, 85 volts rms at 70,000 feet.
Temperature range -65° to +165°C.

REQUIREMENTS

Dimensions and configurations See figures 1, 2, and 3.

Center contact retention
Axial force - 6 pounds minimum.

Force to engage and disengage
Longitudinal force - Not applicable.
Torque - 2 inch-pounds maximum.

Mating characteristics
See figure 2 for SMA dimensions and figure 3 for TNC dimensions.
Center contact (socket)
Oversize test pin Series SMA - .0375+.0001, Series TNC .057 diameter minimum (nonenclosed entry contacts only).
Insertion depth - Series SMA .030/.045, Series TNC .125 minimum.
Number of insertions - Series SMA 3, Series TNC 1.

Insertion force test Steel test pin diameter - Series SMA .0370+.0001, Series TNC - .054 minimum.
Test pin finish - 16 microinches.
Insertion force - Series SMA 3 pounds maximum, Series TNC 2 pounds maximum.

Withdrawal force Steel test pin diameter - Series SMA .0355 -.0001, Series TNC .052 maximum.
Withdrawal force Series SMA 1 ounce minimum, Series TNC 2 ounces minimum
Test pin finish 16 microinches.

Permeability Not to exceed 2

Insulation resistance 5,000 megohms minimum.

Voltage standing wave ratio (VSWR) 1.15±.015 F(GHz) at .5 to 11 GHz.

RF leakage -60 dB minimum tested at a frequency of 2 to 3 GHz.

RF insertion loss .06 $\sqrt{F(GHz)}$ dB maximum at 6 GHz.

Durability 500 cycles minimum at 12 cycles per minute maximum.

Dielectric withstanding voltage 1,000 volts rms minimum at sea level

Contact resistance In milliohms maximum.

	Initial	After environment
Center contact	4.5 1/	6.0
Outer contact	2.2	---

1/ Two center contacts must be mated to the center conductor under test therefore doubling center contact resistance

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Vibration, high frequency Method 204, MIL-STD-202, test condition D, interruptions -1 µs maximum.

Shock (specified pulse) Method 213, MIL-STD-202, test condition I.

Thermal shock Method 107, MIL-STD-202, test condition C.

Moisture resistance 200 megohms minimum

Corona level
  Voltage - 375 volts minimum
  Altitude - 70,000 feet minimum

RF high potential withstanding voltage
  RF voltage - 1,000 volts rms minimum.
  Frequency - 5 to 7.5 MHz.

Salt spray (corrosion). Method 101, MIL-STD-202, test condition B.

Part number M55339/41-50001

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
  Army - EL
  Navy - EC
  Air Force - 85

Review activities
  Army - MI, EL, AT
  Navy - SH
  Air Force - 11, 99
  DLA - ES

User activities
  Army - AT, AR
  Navy - AS, MC
  Air Force - 19

Preparing activity
  Army - EL

Agent
  DLA - ES

(Project 5935-3029-2)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-A-55339/41A ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA(FEMALE) TO SERIES TNC (FEMALE), CLASS 2, STRAIGHT PLUG

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes" in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
	DATE

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REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED.

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